

MAJOR PROJECT

ON

Fake Profile Detection on Social Media Using Machine Learning



GUIDED BY:
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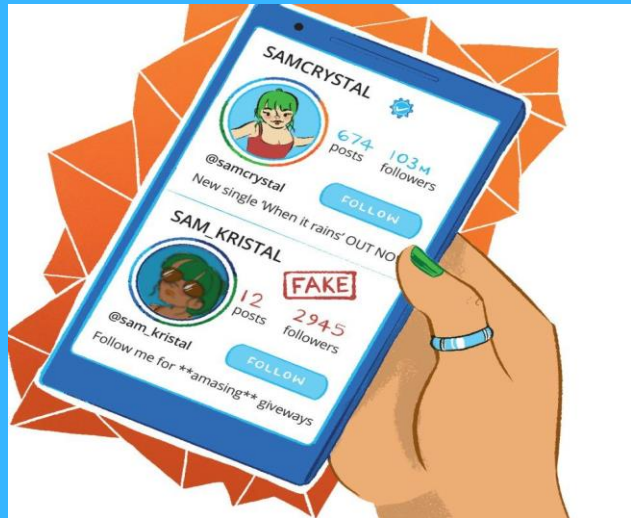
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What are Fake Profiles?

A fake profile is a social media account created using false or stolen identity information. These accounts are often controlled by humans or automated bots and are used for malicious or deceptive activities.

Purpose of Fake Profiles are Spread spam and fake news,Perform phishing and online scams ,Increase fake followers and likes,Identity theft and impersonation

Manipulate public opinion.

Type	Description	Example Use Cases
<u>Bot Accounts</u>	Automated accounts controlled by software	Auto-posting, fake likes, spam campaigns
<u>Imposter Accounts</u>	Pretend to be another person or celebrity	Identity theft, financial fraud, fake celebrity profiles
<u>Spam Accounts</u>	Share advertisements or malicious links	Phishing links, promotional messages, scams
<u>Fake Business Accounts</u>	Fake brands or online stores	Online shopping scams, fake customer support, counterfeit products

- **Our Focus:** Detecting fake social media profiles using Machine Learning (Random Forest) by analyzing profile features such as followers, following, posts, account age, profile picture, and engagement metrics.

LITERATURE REVIEW

Main Author	Research Title	Dataset	Model	Advantages	Limitations	Future Work
Maresh V. et al. (2024)	Machine Learning-Based Fake Profile Detection on Social Networking Websites	Twitter (5,000 profiles)	Random Forest, Decision Tree	High accuracy, easy implementation	Limited to Twitter	Cross-platform detection
B. L. V. S. Aditya et al. (2023)	Heterogenous Social Media Analysis for Efficient Deep Learning Fake-Profile Identification	Multi-platform Dataset	CNN + Deep Transfer Learning	High accuracy using multimodal data	Computationally expensive	Real-time deployment
Giuseppe Sansonetti et al. (2020)	Unreliable Users Detection in Social Media	Twitter Dataset	Deep Learning	Detects unreliable users effectively	Large labeled dataset required	Real-time detection

LITERATURE REVIEW

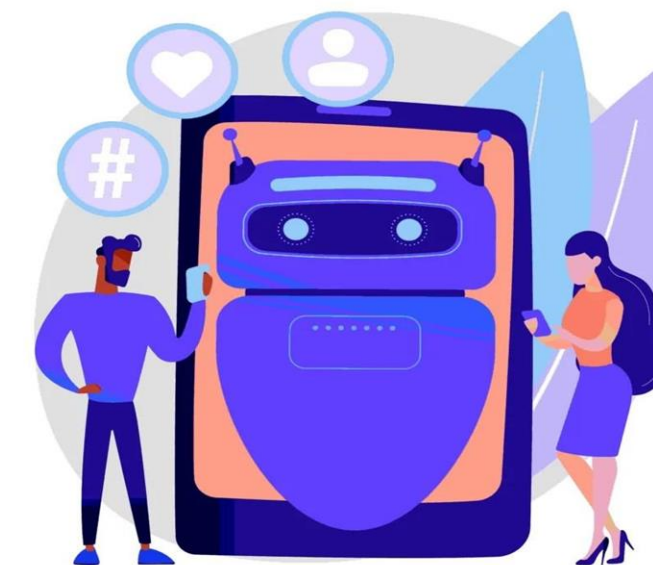
Main Author	Research Title	Dataset	Model	Advantages	Limitations	Future Work
Estee Van Der Walt et al. (2018)	Using Machine Learning to Detect Fake Identities: Bots vs Humans	Social Media Accounts	Random Forest, SVM	Detects bots and fake identities	Handcrafted features	Deep learning-based detection
A. Kumar et al. (2019)	Spammer Detection and Fake User Identification on Social Networks	Twitter Profiles	Random Forest	High precision	Less effective on new attack patterns	Hybrid ensemble learning
Sharma D. & Singh N. (2025)	Fake Profile Detection on Social Networking Sites	Kaggle Fake & Real Users	Random Forest	Fast training and interpretable	Dataset imbalance	Multi-platform AI detection

LITERATURE REVIEW

Main Author	Research Title	Dataset	Model	Advantages	Limitations	Future Work
Ramalingam D. & Valliyammai C. (2018)	Fake Profile Detection Techniques in Large-Scale Online Social Networks	Multiple Datasets	Survey of ML Methods	Comprehensive comparison	Survey only	Hybrid AI models
Goyal B. et al. (2024)	Securing Social Spaces: Fake Profile Detection on Instagram	Instagram Dataset	Random Forest, XGBoost	Better Instagram detection	Platform-specific	Cross-platform monitoring
Terumalasetti S. et al. (2024)	Detecting Malicious Profiles Using Multi-Dimensional Analytics	Multi-dimensional Dataset	Ensemble Learning	High detection accuracy	Complex feature engineering	Explainable AI (XAI)
Susan Mohammed et al. (2024)	Knowledge Rules-Based Decision Tree Classifier Model for Effective Fake Accounts Detection in Social Networks	Twitter (11,737) & Instagram (785)	KRDTC (Decision Tree)	100% accuracy, interpretable IF–THEN rules, real-time detection	Evaluated only on Twitter & Instagram	Facebook, LinkedIn & scalable real-time deployment

PROBLEM STATEMENT

- ❖ Existing The rapid growth of fake profiles on social media has increased spam, scams, and identity theft.
- ❖ Manual detection of fake accounts is difficult due to the large number of users.
- ❖ Existing methods often struggle with evolving fake account behaviors and large-scale datasets.
- ❖ There is a need for an accurate and automated Machine Learning model to classify fake and genuine profiles.



OBJECTIVES

- ❖ *Detect fake social media profiles automatically.*
- ❖ *Classify profiles as Fake or Genuine.*
- ❖ *Develop a Random Forest-based prediction model.*
- ❖ *Improve detection accuracy and reliability. Enhance social media security.*



DATASET & PREPROCESSING DETAILS

Dataset Overview:

- ❑ *Total Profiles: 5,000*
- ❑ *Fake Profiles: 2,500*
- ❑ *Real Profiles: 2500*

Image Specs

- ❑ *Dataset: Twitter Fake & Real User*
- ❑ *Dataset Features: 35 (34 original + Label)*
- ❑ *Target Variable: Label (0 = Genuine, 1 = Fake)*

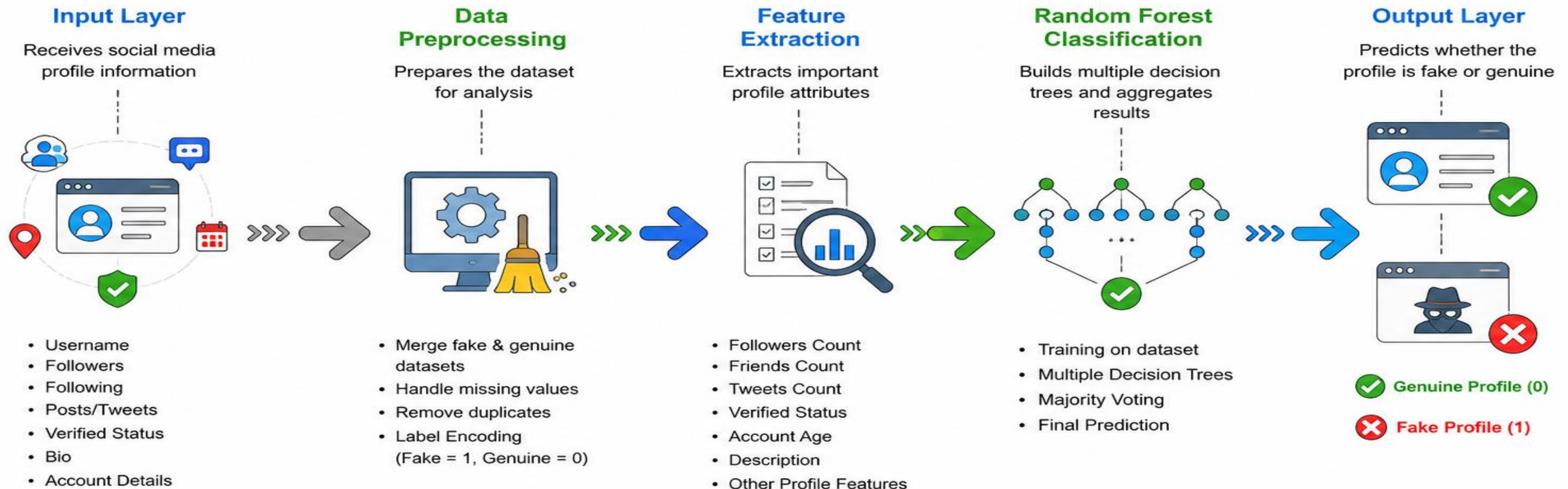
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===== Dataset Summary =====  
Dataset Name      : Twitter Fake & Real User Dataset  
Total Profiles    : 5000  
Fake Profiles     : 2500  
Genuine Profiles  : 2500  
Total Features    : 35  
Target Variable   : Label
```

Preprocessing Steps

- ❑ Merge fake & real datasets
- ❑ Add class labels (Fake = 1, Genuine = 0)
- ❑ Handle missing values
- ❑ Remove duplicate records
- ❑ Feature selection
- ❑ Encode categorical features
- ❑ Train-Test Split (80:20)

MODEL ARCHITECTURE

Fake Profile Detection using Random Forest Classifier



Input Data



Preprocessing



Feature Extraction



Random Forest



Prediction

METHODOLOGY

❑ Data Collection

Collect fake and genuine user datasets.

❑ Data Preprocessing

Merge datasets, clean data, and add labels.

❑ Feature Selection

Select important profile attributes.

❑ Model Training

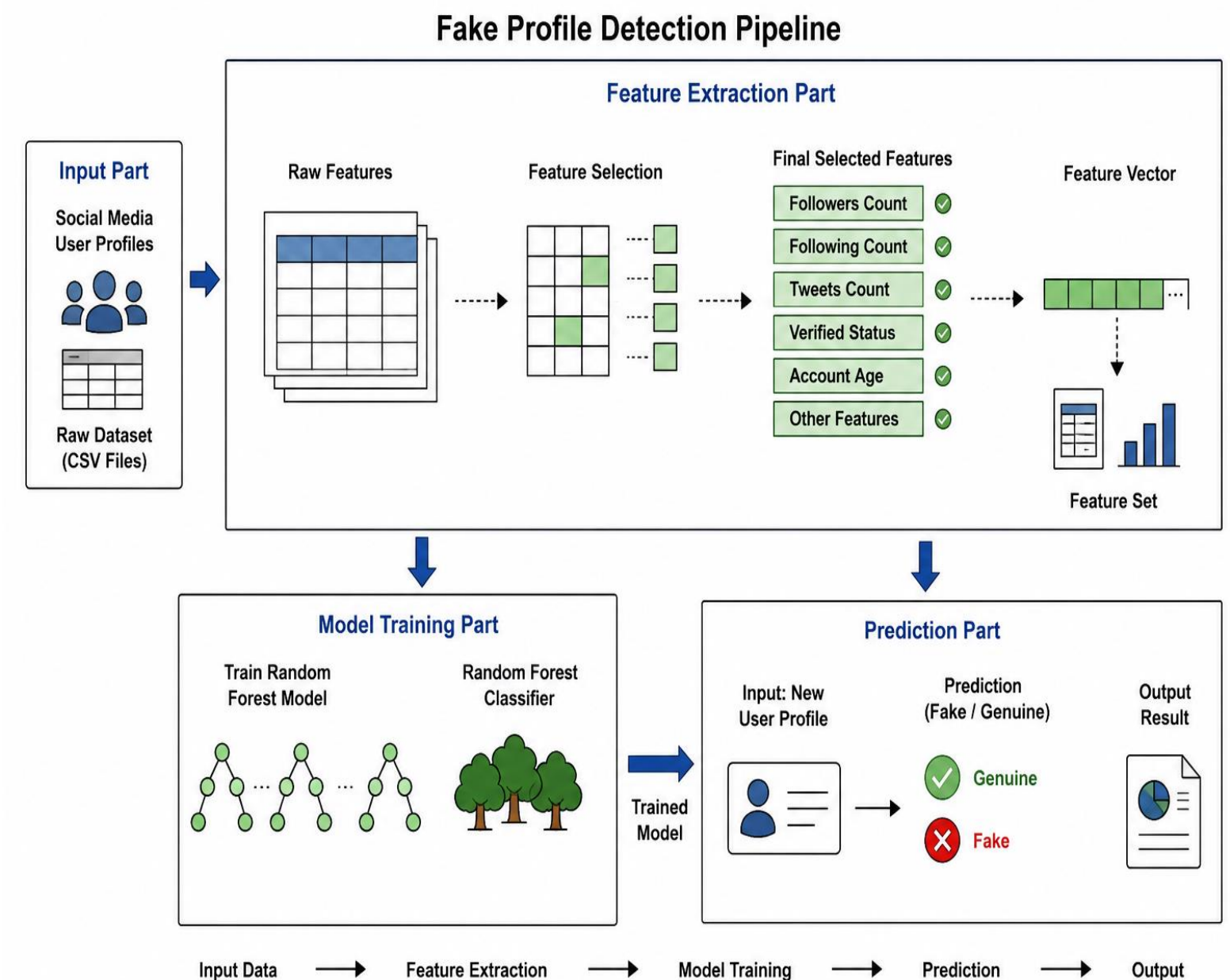
Train the Random Forest classifier.

❑ Model Evaluation

Measure Accuracy, Precision, Recall, and F1-Score.

❑ Profile Prediction

*Classify profiles as **Fake** or **Genuine**.*



What is Random Forest Algorithm?

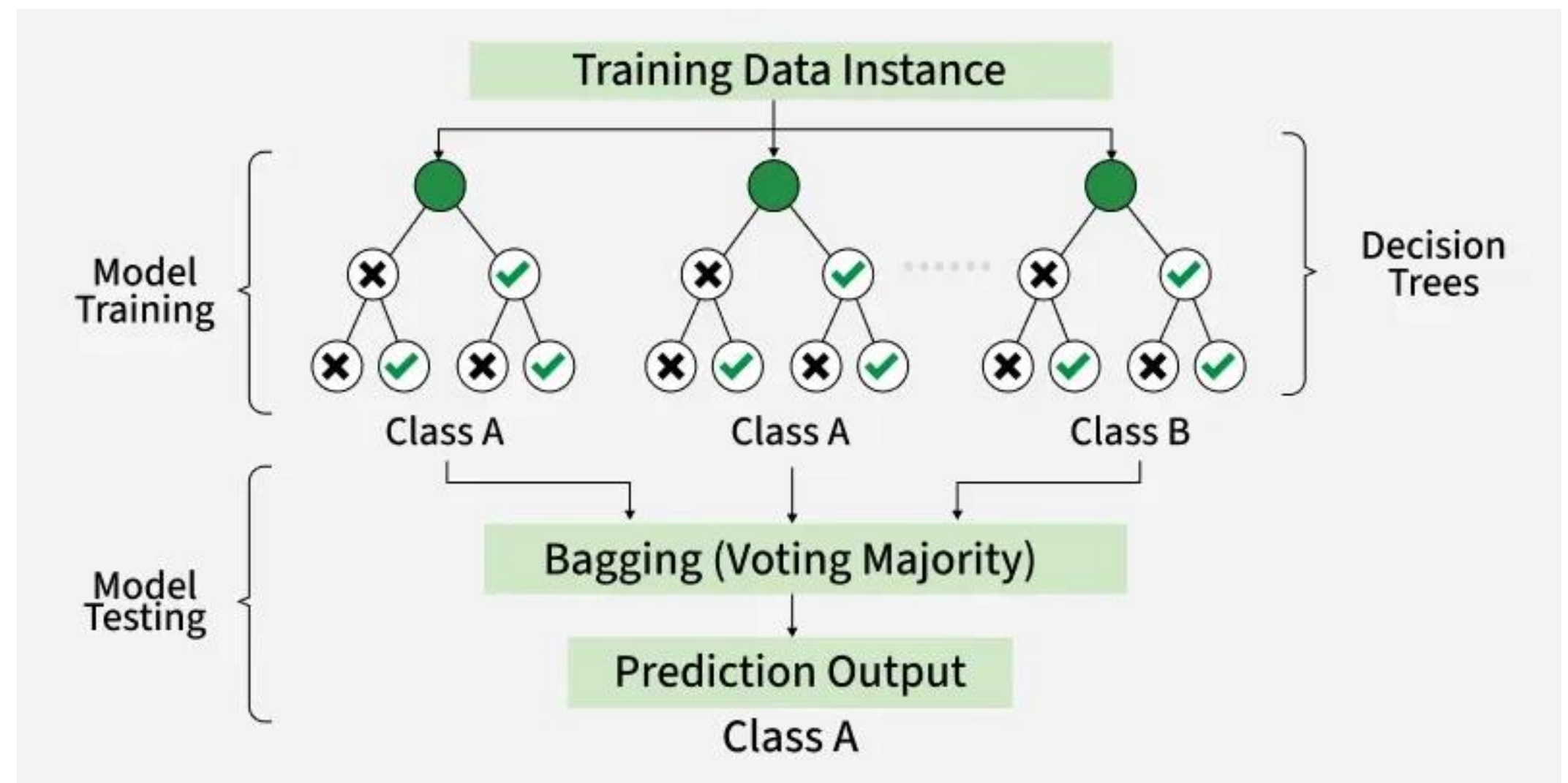
- Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting

- **Working Steps**

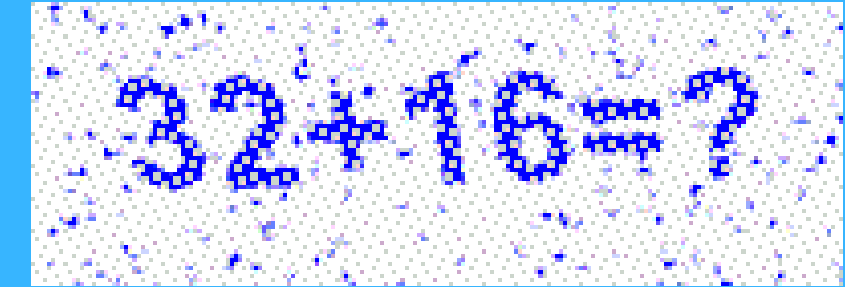
- Create multiple decision trees
- Train each tree using random data samples
- Predict using each tree
- Use **majority voting** for final prediction

- **Advantages**

- High Accuracy
- Handles large datasets
- Reduces overfitting
- Works well with missing value



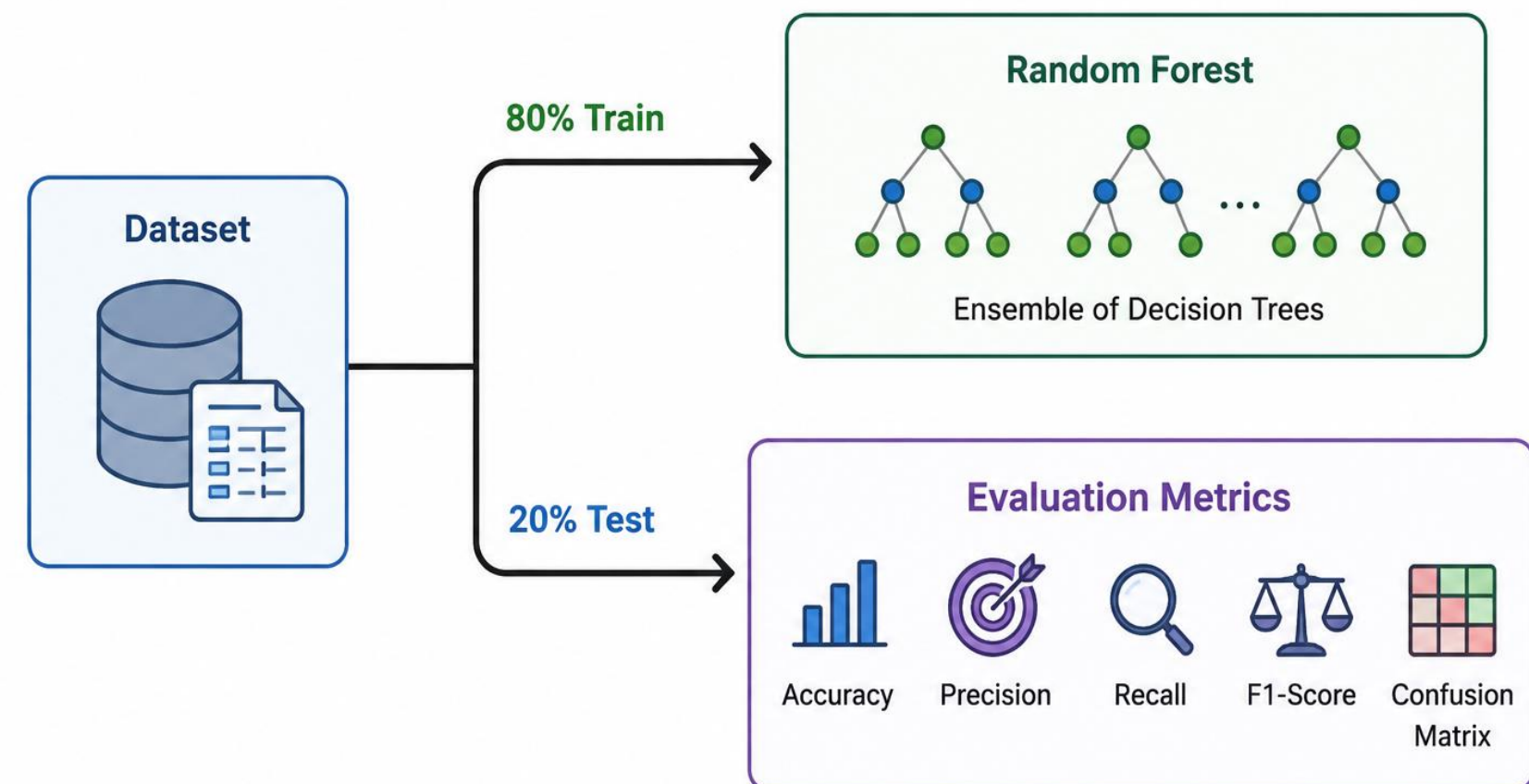
MODEL TRAINING CONFIGURATION



➤ Training Process

- Train-Test Split (80:20)
- Random Forest Training
- Hyperparameter Tuning
- Model Testing

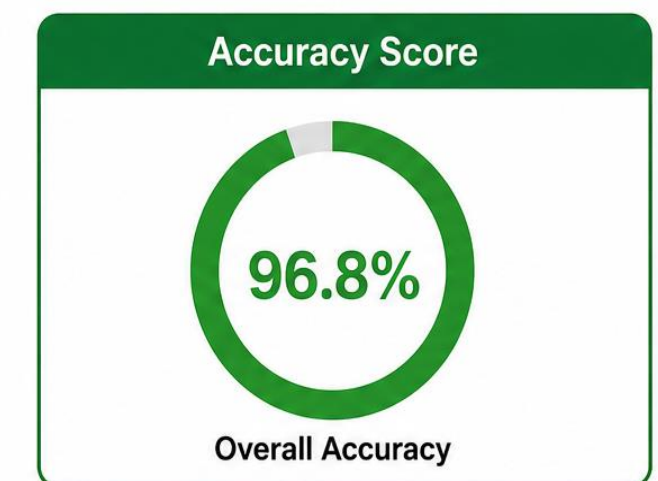
Metric	Purpose
Accuracy	Overall correctness
Precision	Correct fake predictions
Recall	Detect fake profiles
F1-Score	Balance Precision & Recall
Confusion Matrix	Classification performance



RESULTS & PREDICTION

Username	Actual	Predicted
User_01	Fake	Fake
User_02	Genuine	Genuine
User_03	Fake	Fake

Confusion Matrix			
		Predicted	
		Fake (1)	Genuine (0)
Actual	Fake (1)	TP 412	FP 18
	Genuine (0)	FN 21	TN 549

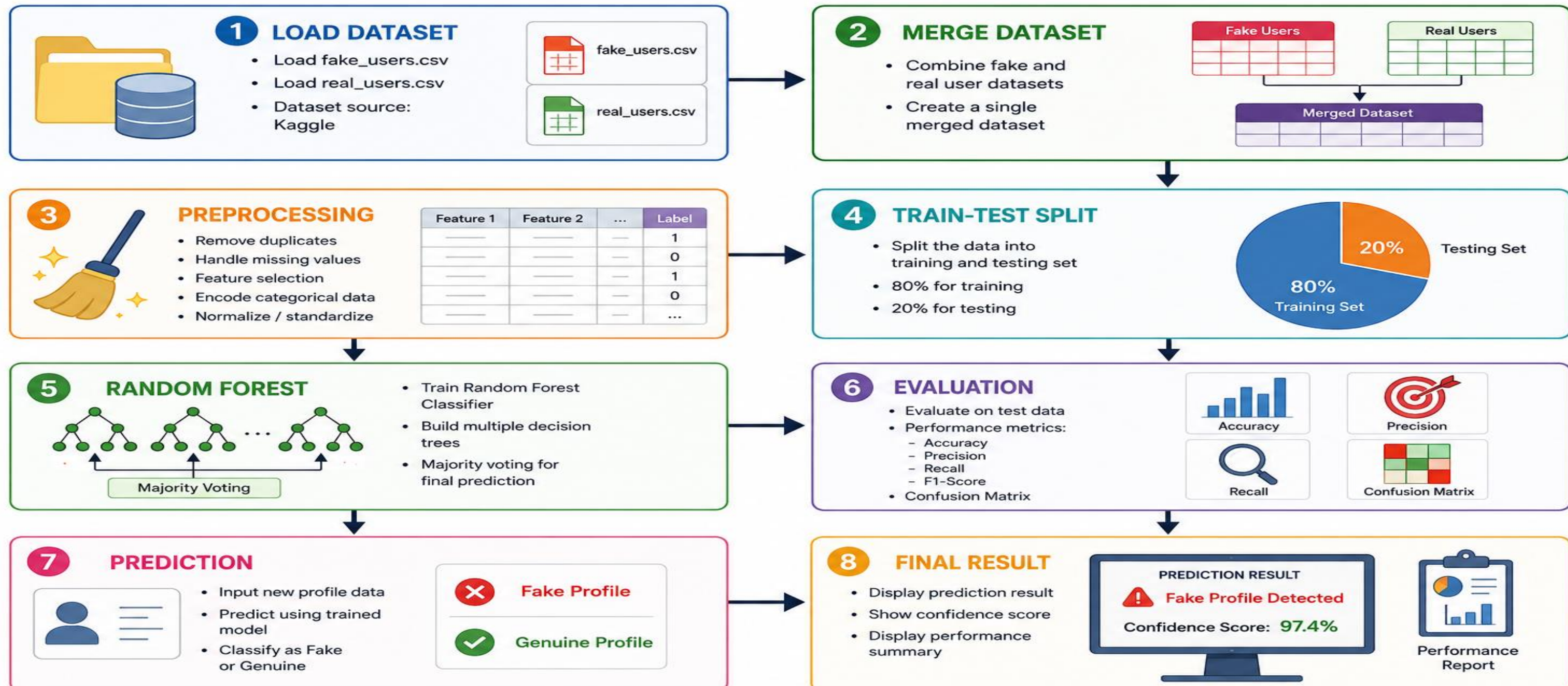


➤ Results

- Fake Profiles Detected
- Genuine Profiles Detected
- Accuracy Score
- Confusion Matrix

Prediction Output		
	 Username	: @john123
	 Followers	: 15
	 Following	: 2500
	 Verified	: No
	 Prediction	:  Fake Profile

WORKING IMPLEMENTATION



LIMITATIONS AND FUTURE WORKS

➤ Limitations

- *Dataset limited to Twitter profiles*
- *Performance depends on data quality*
- *Cannot detect newly evolved fake behaviors*
- *Limited real-time detection capability*

➤ Future Scope

- *Real-time fake profile detection*
- *Support Instagram, Facebook, LinkedIn*
- *Deep Learning models*
- *NLP-based bio analysis*
- *Image verification*
- *Explainable AI (XAI)*



CONCLUSION

- *Developed a machine learning model to detect fake social media profiles.*
- *Used Random Forest for profile classification.*
- *Combined fake and genuine datasets for balanced training.*
- *Achieved reliable classification performance.*
- *The system can assist in improving online security and reducing fake accounts.*



REFERENCES

1. Mahesh, V., Tharun, K., Rushikesh, P., & Saidulu, D. (2024). Machine Learning-Based Fake Profile Detection on Social Networking Websites. *International Journal of Scientific Research in Computer Science, Engineering and Information Technology*.
2. Aditya, B. L. V. S., et al. (2023). Heterogenous Social Media Analysis for Efficient Deep Learning Fake-Profile Identification. *IEEE Access*.
3. Sansonetti, G., et al. (2020). Unreliable Users Detection in Social Media: Deep Learning Techniques for Automatic Detection. *IEEE Access*.
4. Van Der Walt, E., Eloff, J. H. P., & Grobler, J. (2018). Using Machine Learning to Detect Fake Identities: Bots vs Humans. *South African Computer Journal*.
5. Kumar, A., et al. (2019). Spammer Detection and Fake User Identification on Social Networks. *IEEE Access*.
6. Sharma, D., & Singh, N. (2025). A Review of Deep Learning Approaches for Fake Profile Detection on Social Networking Sites. *International Journal of Scientific Research in Science, Engineering and Technology*.
7. Ramalingam, D., & Valliyammai, C. (2018). Fake Profile Detection Techniques in Large-Scale Online Social Network
8. Goyal, B., Gill, N. S., & Gulia, P. (2024). Securing Social Spaces: Machine Learning Techniques for Fake Profile Detection on Instagram. *Social Network Analysis and Mining*,
9. Terumalasetti, S., et al. (2024). Enhancing Social Media User's Trust: A Comprehensive Framework for Detecting Malicious Profiles Using Multi-Dimensional Analytics. *IEEE Access*.
10. Mohammed, S., Al-Aaraji, N., & Al-Saleh, A. (2024). Knowledge Rules-Based Decision Tree Classifier Model for Effective Fake Accounts Detection in Social Networks. *International Journal of Safety and Security Engineering*,